

(a) textbook hard-threshold ABC-PMC: one update per generation

P_{t+1} : the N kept, weighted; the rejected are discarded

barrier: q_{t+1} needs the complete P_{t+1}

$$\epsilon_{t+1} = \text{quantile}_\alpha \{\rho_i\}_{i \leq N}$$

hard: keep $\rho < \epsilon_{t+1}$

$$q_t = \sum_{i \leq N} \omega_i^{(t)} K_{\Sigma_t}(\cdot - \theta_i)$$

$$\omega^{(t+1)} \propto \pi/q_t$$

after acceptance

P_t : $(\theta_i, \rho_i, \omega_i^{(t)})$
gen. $< t$ discarded

$$\Sigma_t = 2 \widehat{\text{Cov}}_{\omega^{(t)}}(\{\theta_i\})$$

weights carried over

$\theta^* \sim q_t$, simulate,
keep if $\rho^* < \epsilon_{t+1}$;
one at a time, until N kept



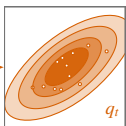
1 state



2 tolerance ϵ



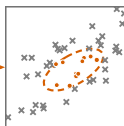
3 weights, Σ



4 proposal



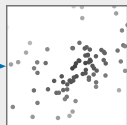
5 draw



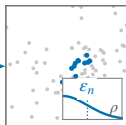
6 weight

(b) generation-free (ours): one update per arrival

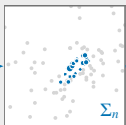
$|\mathcal{H}_n| < k$: prior draw, $w^* = 1$, no ϵ stamp



$\mathcal{H}_n$: all n so far
 $(\theta_i, \rho_i, \epsilon_i, \tau_i, w_i)$

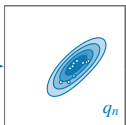


$A_n = \text{Top}_k(\rho)$
 ϵ_n non-increasing, smooth K_{ϵ_n}

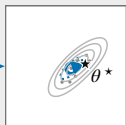


$$\tilde{W}_j \propto w_j K_{\epsilon_n}(\rho_j)$$

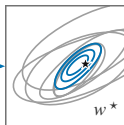
$$\Sigma_n = s \widehat{\text{Cov}}_{\tilde{W}}(A_n)$$



$$q_n = \sum_{j \in A_n} \tilde{W}_j K_{\Sigma_n}(\cdot - \theta_j)$$



$\theta^* = \theta_J + L_n z$, $J \sim \tilde{W}$
one candidate



$$w^* = \pi / \bar{q}_n^{\text{on}} \text{ at } \theta^*$$

θ^* to a free worker; ρ^* arrives later, among other workers' results: append $(\theta^*, \rho^*, \epsilon_{\tau^*}, \tau^*, w^*)$, $\tau^* = n$